

Wireless Communication

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MIMO (Multiple Input Multiple Output)

- MIMO uses multiple antennas at both transmitter and receiver.
- Instead of sending one signal, multiple signals are transmitted simultaneously.
- This improves:
 - ▶ Data rate (through spatial multiplexing)
 - ▶ Reliability (through diversity)
- Basic idea: The received signal expression in MIMO becomes

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

where

- ▶ $\mathbf{x} \in \mathbb{C}^{N_t \times 1}$: Transmit signal vector
- ▶ $\mathbf{y} \in \mathbb{C}^{N_r \times 1}$: Received signal vector
- ▶ $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$: Channel matrix
- ▶ $\mathbf{n} \in \mathbb{C}^{N_r \times 1}$: Noise vector

- MIMO converts the wireless channel from a limitation into an opportunity by exploiting multiple parallel spatial channels.
- The MIMO equation for $N_t = 3, N_r = 3$ is given by

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

where

h_{ij} : Channel from j^{th} transmit antenna to i^{th} receive antenna

- From the above matrix and vector representation, we can write the expressions for received signal y_1, y_2, y_3 as

$$y_1 = h_{11}x_1 + h_{12}x_2 + h_{13}x_3 + n_1$$

$$y_2 = h_{21}x_1 + h_{22}x_2 + h_{23}x_3 + n_2$$

$$y_3 = h_{31}x_1 + h_{32}x_2 + h_{33}x_3 + n_3$$

- Thus, we can see that each received signal is a combination of all transmitted signals.
- So, MIMO creates mixing at the channel, and the receiver's job is to unmix it.

MIMO Channel Capacity

- Capacity of MIMO system is given by:

$$C = \log_2 \det \left(\mathbf{I} + \frac{\rho}{N_t} \mathbf{H} \mathbf{H}^H \right)$$

where

- ▶ ρ : Signal-to-noise ratio (SNR)
 - ▶ N_t : Number of transmit antennas
 - ▶ $\mathbf{H}$: Channel matrix
 - ▶ $\mathbf{H}^H$: Hermitian (conjugate transpose)
- Capacity increases with number of antennas.
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 - For a Single Input Single Output (SISO) system, the capacity is:

$$C = \log_2(1 + \rho)$$

- Thus in SISO, capacity depends only on SNR.

- Numericals in MIMO channel capacity: Ref Class Notes.
- From the numericals, we have seen that in MIMO, channel capacity depends on $\mathbf{H}$, so better the $\mathbf{H}$ better is channel capacity.
- Thus in MIMO, next question comes to mind is can we transform $\mathbf{H}$ into better form ?

Signal Mixing in MIMO

- In MIMO, received signals are combinations of transmitted signals:

$$y_1 = h_{11}x_1 + h_{12}x_2 + n_1$$

$$y_2 = h_{21}x_1 + h_{22}x_2 + n_2$$

- Each received signal depends on multiple transmitted signals.
- This causes **interference between data streams**.

Goal: Remove Interference

- We want to convert the system into independent channels:

$$y'_1 = \sigma_1 x'_1 + n'_1$$

$$y'_2 = \sigma_2 x'_2 + n'_2$$

where σ_1, σ_2 are clean channel gains (channel coefficients).

- Now each output depends on only one input.
- No interference between signals.

Solution: Singular Value Decomposition (SVD)

- The MIMO channel matrix can be decomposed as:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

- $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices. Unitary matrices only rotate signals, they don't amplify or attenuate them.
- $\mathbf{\Sigma}$ is a diagonal matrix.
- This transformation converts the MIMO channel into parallel independent channels.

Applying SVD to MIMO System

- Original MIMO system:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

- Substitute $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$:

$$\mathbf{y} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \mathbf{n}$$

- Define new transmit signal:

$$\mathbf{x}' = \mathbf{V}^H\mathbf{x}$$

- Multiply both sides by $\mathbf{U}^H$:

$$\mathbf{U}^H\mathbf{y} = \mathbf{\Sigma}\mathbf{x}' + \mathbf{U}^H\mathbf{n}$$

- Define:

$$\mathbf{y}' = \mathbf{U}^H\mathbf{y}, \quad \mathbf{n}' = \mathbf{U}^H\mathbf{n}$$

- Final system:

$$\mathbf{y}' = \mathbf{\Sigma}\mathbf{x}' + \mathbf{n}'$$

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

$$\begin{bmatrix} y'_1 \\ y'_2 \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} x'_1 \\ x'_2 \end{bmatrix} + \begin{bmatrix} n'_1 \\ n'_2 \end{bmatrix}$$

$$y'_1 = \sigma_1 x'_1 + n'_1$$

$$y'_2 = \sigma_2 x'_2 + n'_2$$

- Earlier, each received signal was a mixture of all transmitted signals.
- After SVD, each output depends on only one input $\rightarrow$ no interference.
- The MIMO channel is converted into independent parallel channels.

Limitation of SVD

- SVD converts the MIMO channel into parallel independent channels.
- However, it requires **Channel State Information (CSI)** at the transmitter.
- In practical systems, transmitter often does not know the channel.
- Therefore, SVD-based transmission is not always feasible.

Need for Practical Detection Techniques

- Without SVD preprocessing, received signals remain mixed:

$$y_1 = h_{11}x_1 + h_{12}x_2 + n_1$$

$$y_2 = h_{21}x_1 + h_{22}x_2 + n_2$$

- We need methods to separate these signals at the receiver.
- One such technique is **V-BLAST**((Vertical Bell Labs Layered Space-Time)).

How V-BLAST Works

- V-BLAST separates transmitted signals at the receiver step-by-step.
- Step 1: Detect the strongest signal first.
- Step 2: Subtract its effect from the received signal.
- Step 3: Detect the next strongest signal.
- Step 4: Repeat until all signals are detected.
- This process is called Successive Interference Cancellation (SIC).

V-BLAST Numerical Example

Question:

Consider a 2×2 MIMO system with

$$\mathbf{H} = \begin{bmatrix} 1 & 0.5 \\ 0.2 & 1 \end{bmatrix}$$

And the transmitted signal vector is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

If the received values are:

$$y_1 = 2, \quad y_2 = 2.2$$

Using V-BLAST (Successive Interference Cancellation), estimate the transmitted signals x_1 and x_2 .

Sol: Ref Class Notes

Alamouti Scheme

- Used to improve reliability (reduce errors).
- Instead of sending different signals, we send same information in a smart way.
- Uses 2 transmit antennas and 2 time slots.
- Time 1:
 - ▶ Antenna 1 sends s_1
 - ▶ Antenna 2 sends s_2
- Time 2:
 - ▶ Antenna 1 sends $-s_2^*$
 - ▶ Antenna 2 sends s_1^*
- So receiver gets multiple versions of same signal.

Alamouti Example (With Fading)

Given:

- Symbols: $s_1 = 1$, $s_2 = 2$
- Channel: $h_1 = 0.1$ (weak), $h_2 = 1$ (strong)

Received Signals:

$$y_1 = h_1 s_1 + h_2 s_2 = 0.1(1) + 1(2) = 2.1$$

$$y_2 = h_1(-s_2) + h_2(s_1) = 0.1(-2) + 1(1) = 0.8$$

- First path ($h_1 = 0.1$) is very weak $\rightarrow$ signal almost lost.
- But second path ($h_2 = 1$) is strong $\rightarrow$ signal survives.
- So even though one path failed, the other carries useful information.
- Receiver uses both signals to recover s_1 and s_2 .
- This is called **diversity gain**.